



THE SUPERIOR UNIVERSITY LAHORE

Quiz - 1 - Solution

Semester: 3rd

Section: C

Session: Fall 2025

Faculty of Computer Science and Information Technology

Subject: Data Structures and Algorithms

QCH:

Total Marks: 10

Name.....

Roll No.....

Instructions:

1. Cheating is not allowed, otherwise 0 marks will be rewarded.

Task #	CLO #	Domain and BT Level	Total Points
1	1	C6	05
2	1	C3	05

Task 1: Prioritizing Tasks in a Queue (Incorporating Queue)

(5 Marks)

```
undo_stack = []
redo_stack = []
def do_action(action):
    undo_stack.append(action)
    redo_stack.clear() # Clear redo stack on new action
    print(f"Action performed: {action}")
def undo():
    if undo_stack:
        action = undo_stack.pop()
        redo_stack.append(action)
        print(f"Undo: {action} undone")
    else:
        print("Nothing to undo.")
def redo():
    if redo_stack:
        action = redo_stack.pop()
        undo_stack.append(action)
        print(f"Redo: {action} redone")
    else:
        print("Nothing to redo.")
def view_stacks():
    print("Undo Stack:", undo_stack)
    print("Redo Stack:", redo_stack)
do_action("Type Hi")
do_action("Add !")
view_stacks()
undo() # Undo last action
view_stacks()
```

Task 2: Call Center Simulation

(5 Marks)

```
from collections import deque

# Task 2: Call Center Simulation using Queue
call_queue = deque()

def enqueue_call(name):
    call_queue.append(name)
    print(f"{name} added to the queue.")

def dequeue_call():
    if call_queue:
        served = call_queue.popleft()
        print(f"{served} has been served and removed from the queue.")
    else:
        print("No callers in the queue.")

def peek_call():
    if call_queue:
        print(f"Next caller to be served: {call_queue[0]}")
    else:
        print("No callers in the queue.")

def display_queue():
    if call_queue:
        print("Current callers in queue:", list(call_queue))
    else:
        print("No callers waiting.")

# ----- Simulation -----
enqueue_call("Ali")
enqueue_call("Sara")
enqueue_call("Ahmed")

display_queue()
peek_call()

dequeue_call()
display_queue()
```